

ALI LENJANI

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Skills / Tech Stack

Core Strengths: Recommender systems (two-tower, vector search), model-based RL for LTV optimization, multimodal machine learning, computer vision, digital twins & simulation, human-in-the-loop evaluation, large Scale Distributed Training.

Data Science & ML: Python, R, SQL, PyTorch, TensorFlow, Keras, Scikit-Learn, PySpark, Pandas, NumPy, SciPy, OpenCV, Statsmodels.

MLOps, Serving & Retrieval: ONNX Runtime, TensorRT, MLflow, MosaicML Streaming, Vector Databases, open-source RAG pipelines, vLLM, Hugging Face, LangChain, Grafana, Locust.

Cloud & Data Platforms: GCP, Azure, AWS, Databricks, Snowflake.

Product & Stakeholder Management: Cross-functional squad leadership, ML roadmap development, business KPI translation (growth/loyalty), A/B test design & readouts, Agile/Scrum.

Optimization & Modeling: Google OR-Tools, SciPy Optimize, PyMC (Bayesian modeling), stochastic simulation & Monte Carlo analysis.

Developer Tools & Productivity: GitHub, GitHub Copilot, Codex, Claud Code, Open Claw, Cursor, Replit, Jira, L^AT_EX.

EXPERIENCE

Lead ML/Data Scientist

Albertsons Companies Inc

Aug 2022 – Present

San Francisco, CA

- End-to-End Real-Time Recommendation Platform (15M+ Users):
 - Built and productionized a latency-critical two-tower recommendation system serving 15M+ active users and trained on 1.5B+ interactions.
 - Designed a sub-10ms retrieval pipeline by indexing 500k+ items in Azure AI Search (HNSW). Optimized User Tower inference utilizing ONNX Runtime, achieving a 3–5x speedup over native PyTorch.
 - Designed a Ranking-Aware Contrastive Loss function combining InfoNCE with Hard Negative Mining (same-taxonomy sampling) to force fine-grained attribute discrimination.
 - Overcame data bottlenecks by implementing MosaicML Streaming for sharded reads across 8x NVIDIA A100 GPUs. Engineered a bi-weekly warm-start training strategy that reduced compute costs by 5x.
- LLM-as-a-Judge for Recommendation Systems: Designed and developed a multi-dimensional evaluation platform to quantify semantic complementarity. Engineered a declarative "Human-Proxy" judging pipeline that scores 7 orthogonal latent dimensions using Chain-of-Thought (CoT) reasoning to automate quality gating.
- Autonomous Agent Simulation: Built a high-fidelity Counterfactual Offline Simulation Platform integrated with LLM agents. Features include "time-travel" historical sampling and self-adaptive metric tuning to minimize the sim-to-real gap for deployment decisions.
- Vision AI for Manufacturing Quality Control:

- Architected an end-to-end computer vision inspection framework across multiple CTQ checkpoints for automated defect detection and station health monitoring.
- **Object Detection & Edge Deployment:** Trained multi-class object detection models (PyTorch) for defect identification. Engineered operational decision logic, optimizing inference thresholds to minimize false positives in high-throughput environments.
- **Data Pipelines & Hard Negatives:** Built a leakage-safe data workflow using GCP/GCS and Label Studio. Overcame physical manufacturing constraints (lighting, misalignment, class imbalance) via targeted hard negative mining.

Post-doctoral Research Scholar

2021 – 2022

Stanford University

Palo Alto, CA

- Multimodal Machine Learning: Designed and developed an automated data pipeline and deep learning-based predictive model for monitoring critical health indicators, targeting a significant reduction in field survey workload.

Post-doctoral Research Associate (Space Habitat Digital Twin)

2020 – 2021

NASA STRI, Purdue University

West Lafayette, IN

- Physics-Informed Digital Twin: Spearheaded the CDCM platform, integrating thermodynamic and structural models to simulate deep-space habitat resilience.
- Thermal-Mechanical Simulation: Modeled coupled interactions between active thermal control, power generation, and structural integrity under stochastic disturbances (e.g., micrometeoroid impacts, thermal cycling).
- Resilience-Based Trade Studies: Executed large-scale Monte Carlo simulations to optimize design configurations against stakeholder value metrics (NPV & Risk), enabling risk-informed mission planning.

Graduate Research Assistant (Time-Critical Decision Support)

2016 – 2020

Purdue University

West Lafayette, IN

- Developed data pipelines for automated collection and pre-processing of Streetview 360-degree images for post-event building evaluation (using Faster R-CNN and depth detection).
- Created ML-based exposure and vulnerability models for assets in the built environment using computer vision and image fusion techniques.
- Developed a risk-informed resource allocation algorithm to minimize post-event inspection time and improve community resilience.

EDUCATION

Post-doctoral Research Scholar, Human-Centered AI

2021–2022

Stanford University

Palo Alto, CA, USA

- **Research Focus:** Multimodal Machine Learning in Global Health Monitoring and Resource Allocation.

Post-doctoral Research Associate, Complex Systems / Digital Twin

2020–2021

NASA STRI, Purdue University

West Lafayette, IN, USA

- **Research Focus:** AI-based platform for risk-informed project portfolio optimization for complex systems design (Space Habitat Digital Twin).

Ph.D., Mechanical Engineering

2016–2020

Purdue University

West Lafayette, IN, USA

- **Dissertation:** AI-based Platform for Risk-Informed Decision Making in Socio-Technical Systems (Deep Learning, Computer Vision, and Stochastic System Simulation).

M.Sc., Mechanical Engineering

2015–2016

Northern Illinois University (NIU)

DeKalb, IL, USA

- **Thesis:** Reinforcement Learning and Differential Dynamic Programming for 2D dynamics simulation (Spumone).

B.Sc., Mechanical Engineering and Computational Sciences

Iran University of Science and Technology (IUST)

2005–2011

Tehran, Iran

SELECTED PEER-REVIEWED JOURNAL PUBLICATIONS

[1] **Lenjani, A.**, Dyke, S. J., Bilonis, I., Yeum, C. M., Kamiya, K., Choi, J., Liu, X. (2020). Towards fully automated post-event data collection and analysis: Pre-event and post-event information fusion. *Engineering Structures*, 208, 109884.

[2] **Lenjani, A.**, Bilonis, I., Dyke, S. J., Yeum, C. M., Monteiro, R. (2020). A resilience-based method for prioritizing post-event building inspections. *Natural Hazards*, 100(2), 877-896.

[3] **Lenjani, A.**, Yeum, C. M., Dyke, S., Bilonis, I. (2019). Automated building image extraction from 360° panoramas for postdisaster evaluation. *Computer-Aided Civil and Infrastructure Engineering*, 51.

CONTRIBUTION TO GRANTS AND FUNDING

Resilient ExtraTerrestrial Habitats Institute (RETHi) | NASA

Developed autonomous systems for self-monitoring and adaptation to ensure habitat sustainability in deep-space environments. [\[Link\]](#)

Automating Vulnerability Quantification using Street-View Images | NSF

Applied computer vision to automate the exposure and vulnerability quantification of residential assets for disaster resilience. [\[Link\]](#)

Integrating Human and Machine for Post-Disaster Visual Data Analytics | NSF

Integrated human and machine intelligence via modern media tools to create efficient, accurate disaster response systems. [\[Link\]](#)

EAGER: Active Citizen Engagement to Enable Lifecycle Management | NSF

Promoted active citizen engagement in data collection and decision-making to ensure better management of critical infrastructure. [\[Link\]](#)

CDS&E: Time-Critical Decision-Support for Disaster Response | NSF

Automated visual analytics for structural engineering to support time-critical decisions during and after disaster events. [\[Link\]](#)

Work Authorization

U.S. Permanent Resident (Green Card Holder)